# Harshith Kadaru

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## EDUCATION

**The George Washington University** — Washington, DC
*Master of Science, Data Science, GPA: 4.0* — Expected May 2027

**CVR College of Engineering** — Hyderabad, India
*Bachelor of Technology, Computer Science and Information Technology* — April 2025

## SKILLS

**Languages:** Python, SQL, R, TypeScript, JavaScript
**ML and Statistics:** scikit-learn, LightGBM, PyTorch, pandas, SciPy, time series forecasting, A/B testing, power analysis
**AI and LLMs:** LLM evaluation, LLM-as-a-judge, agent harnesses, MCP servers, tool calling, RAG, embeddings, ChromaDB
**Engineering:** FastAPI, React, Redis, MySQL, MongoDB, Docker, GCP (Cloud Run, Cloud SQL), GitHub Actions CI

## EXPERIENCE

**MATClinics (SUDx Analytics)** — Maryland (Hybrid)
*Software Engineering Intern* — Jun 2026 - Aug 2026

- Migrated two production ML scoring pipelines from R / H2O / Java to Python, with 100% and 97.3% parity.
- Root-caused 5 months of silent job failures, backfilled the missing scores, and made the monthly rebuild idempotent.
- Moved real-time updates from in-memory WebSockets to Redis pub/sub to work across scaled Cloud Run instances.
- Built Python ETL into MySQL (Cloud SQL) and FastAPI telehealth services (Google Meet, Twilio SMS) with audit logging.
- Shipped 15+ React / TypeScript features to a multi-site clinical dashboard from direct provider feedback.

**The George Washington University** — Washington, DC
*Graduate Instructional Assistant, DATS 1001: Data Science for All* — Aug 2026 - Present

- Lead labs, grading, and office hours for R-based data wrangling, visualization, and statistical inference.

## PROJECTS

**claimcheck | Python, Docker, GitHub Actions | GitHub** — Oct 2026
*Open-source benchmark testing whether AI coding agents falsely claim success*

- Built 205 ground-truth tasks and a grader no cheat got past (0 of 1,236 attempts); ran 530 GPT-OSS runs at $0.
- Without running tests, models falsely claimed success on 57-60% of unsolvable tasks; with tests, 14-22% ($p < 0.002$).
- Pre-registered McNemar and Fisher tests; a judge from another model family matched ground truth on 96% (kappa 0.90).
- Fine-tuned DistilBERT in PyTorch to flag false success claims: 91% accuracy in 5-fold CV split by task, at $0.

**ML Model Service with Drift Monitoring | Python, LightGBM, FastAPI, Docker | GitHub** — Oct 2026

- Served a LightGBM hotel cancellation model (ROC AUC 0.878) with FastAPI in Docker, logging predictions to SQLite.
- Replayed 55,095 later bookings: AUC fell to 0.76, but drift alarms did no better than random (precision 0.29).
- Traced the drop to a shift in booking mix (Spearman -0.84) and measured detector power and false alarms by simulation.

**claimcheck-mcp | Python, MCP, Docker | GitHub** — Oct 2026

- MCP server giving AI agents 3 tools over the 205 tasks in a Docker sandbox; fixed a symlink leak of answer files.

**A/B Experimentation and Decision Framework | Python, SciPy, pandas | GitHub** — Personal

- Built SRM checks, z-tests, Welch's t-tests, and guardrails that decide ship, continue, or rollback for each experiment.
- Showed with 2,000 A/A simulations that peeking raised false positives from 5% to 19%; O'Brien-Fleming restored 5%.
- Added power analysis (sample size, achieved power, test duration) validated against statsmodels, with 92 tests in CI.

**Demand Forecasting with Backtesting | Python, LightGBM, statsmodels | GitHub** — Oct 2026

- Forecast weekly sales for 3,049 Walmart (M5) items with ETS and LightGBM using lag and calendar features.
- 6-fold rolling-origin backtests with leakage tests: LightGBM MASE 0.963 vs seasonal naive 1.437.

**RiskBot | Python, FastAPI, ChromaDB, GPT-OSS 120B | GitHub | Live Demo** — Dec 2025

- Flags sensitive HR and personal data with checksum rules and GPT-OSS 120B: F1 0.97 on a held-out synthetic set.

## ACHIEVEMENTS

- **Winner, GW Data Science Association Datathon 2026:** built a 24-page Python and Streamlit platform analyzing 500K+ rows, using K-Means clustering, regression, and simulation to recommend policy interventions.